

CS23532 – COMPUTER NETWORKS LAB MANUAL

Practical–2

Aim

To study and understand different types of network cables used in computer networking.

Objective

To identify various types of network cables, their categories, transmission speeds, advantages, disadvantages, and applications.

Types of Network Cables

The different types of cables commonly used in networking are:

- **Unshielded Twisted Pair (UTP) Cable**
- **Shielded Twisted Pair (STP) Cable**
- **Coaxial Cable**
- **Fibre Optic Cable**

Cable Type	Category / Type	Maximum Data Transmission	Advantages	Disadvantages	Applications / Uses
UTP	Cat 3	Up to 10 Mbps	Low cost, easy to install, lightweight	More susceptible to EMI and noise	10BASE-T Ethernet, telephone networks
UTP	Cat 5	Up to 100 Mbps	Inexpensive, easy installation	More interference than shielded cables	Fast Ethernet
UTP	Cat 5e	Up to 1 Gbps	Affordable, widely available, supports Gigabit Ethernet	Less protection against interference	Fast Ethernet, Gigabit Ethernet
UTP	Cat 6	Up to 10 Gbps	Higher bandwidth, better performance	More expensive than Cat 5e	Gigabit Ethernet, 10G Ethernet
STP	Cat 6 / 6a	Up to 10 Gbps	Shielded, less susceptible to noise and interference, high performance	Expensive, difficult to install	Gigabit Ethernet, 10G Ethernet, data centres
SSTP	Cat 7	Up to 10 Gbps	Excellent shielding, high bandwidth, reduced interference	Expensive, requires proper installation	Gigabit Ethernet, 10G Ethernet
Coaxial Cable	RG-6, RG-59, RG-11	Up to 10–100 Mbps	High bandwidth, good resistance to interference, low signal loss	Bulky, limited distance, comparatively expensive	Television networks, broadband Internet
Fibre Optic Cable	Single-mode	Very high speed	High bandwidth, long-distance transmission, high security, immune to EMI	Expensive, requires skilled installation	WANs, backbone networks, data centres
Fibre Optic Cable	Multi-mode	Up to 100 Gbps	High speed, high bandwidth, reliable transmission	More limited distance than single-mode, expensive	LANs, data centres, campus networks

- * Actual speed depends on cable length and network equipment.
- * Speed depends on the specific coaxial standard and equipment.
- * Depends on the fibre type, transceivers, distance, and networking standard

1. Unshielded Twisted Pair (UTP) Cable

UTP cable consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference. It is one of the most commonly used cables in Local Area Networks (LANs).

Advantages:

- Low cost
- Easy to install
- Lightweight and flexible
- Widely available

Disadvantages:

- More susceptible to electromagnetic interference (EMI)
- More affected by external noise compared with shielded cables
- Limited transmission distance

Applications:

- Ethernet networks
- Computer-to-switch connections
- Telephone networks
- LAN installations

2. Shielded Twisted Pair (STP) Cable

STP cable is similar to UTP but includes additional shielding around the twisted pairs. The shielding provides better protection against electromagnetic interference and noise.

Advantages:

- Better protection from EMI
- Less susceptible to noise
- Higher reliability in electrically noisy environments
- Supports high-speed networking

Disadvantages:

- More expensive than UTP
- More difficult to install
- Requires proper grounding

Applications:

- Gigabit Ethernet
- 10 Gigabit Ethernet
- Industrial networks
- Data centres

3. Coaxial Cable

Coaxial cable contains a central copper conductor, an insulating layer, a metallic shield, and an outer protective covering. It provides good protection against electromagnetic interference.

Advantages:

- Good bandwidth
- Resistant to interference
- Durable
- Relatively low signal loss

Disadvantages:

- Bulky compared with twisted-pair cables
- Less flexible
- Limited transmission distance depending on the standard
- Installation can be more difficult

Applications:

- Cable television
- Broadband Internet
- CCTV systems
- Older Ethernet networks

4. Fibre Optic Cable

Fibre optic cable transmits data using pulses of light through optical fibres instead of electrical signals. It provides very high bandwidth and is suitable for long-distance communication.

Types:

1. **Single-mode fibre**
2. **Multi-mode fibre**

Advantages:

- Very high transmission speed
- High bandwidth
- Long-distance transmission
- Immune to electromagnetic interference
- High security
- Low signal attenuation

Disadvantages:

- Higher installation cost
- Requires skilled technicians
- Fibre can be more fragile than copper cable
- Special equipment is required for termination and testing

Applications:

- Internet backbone networks
- Wide Area Networks (WANs)
- Data centres
- Long-distance communication
- High-speed LANs

Result

Thus, the different types of network cables such as **UTP, STP, Coaxial and Fibre Optic cables** were studied along with their types, transmission speeds, advantages, disadvantages, and applications.